1. Lift-dense factor bases: twist avoidance through the trace

Study `research/lift_dense_factor_base_20261027/`, 2026-10-27.

- Run: `reports/run-20261027T0356Z/`, master seed `0x11F7_2026_1027_C`.
- Protocol: `PROTOCOL.md`, frozen at `80ac6c421` before the run. Amendments, all made after the run:
  - 1 (`397f055bf`): empirical variance;
  - 2 (`ce62b2a9e`): rustfmt only, with byte-identical outputs;
  - 3 (`0e40561a6`): review fixes.
- Harness: `src/main.rs`, std-only Rust.
- Real-parameter check: `src/bin/c2pnb.rs`, native.
- Scoring: `summarize/`, native.
- Plotting: `plot.py`, layout only.
- Historical Python scripts (`c2pnb/*.py`, `summarize.py`) are kept as evidence and no longer executed.
- Lean: `lean/LiftDense.lean`, with the proof map in `lean/PROOF_MAP.md`. Theorem 2 is proved and audited; see §Lean.

Each statement is tagged **theorem**, **exact computation**, **measured**, **model output** or **not established**.

1.1. Question

On $E : y^2 + xy = x^3 + ax^2 + b$ over $\mathbb{F}_{2^N}$, a nonzero x lifts to $E(\mathbb{F}_q)$ iff $\mathrm{Tr}(x + a + b/x^2) = 0$; otherwise it lifts to the quadratic twist.

The earlier twist-obstruction study showed that twist points add no relations. So in a factor-base subspace V, the twist-lifting abscissae are dead weight. This study asks whether V can avoid the twist, and what the trace homomorphism, used as a membership condition rather than a sum constraint, buys.

1.2. Statements and their status

1. **Yield law (theorem).** Every ordered m-tuple of the factor base F sums to exactly one point. So the mean number of ordered decompositions of a target is $|F|^m / \#E$, with $|F| = 2|V \cap L|$. A subspace with lift density ρ yields $(2\rho)^m$ times a random subspace of the same dimension.
   - **Measured:** 87 rows of point-arithmetic counts, $m = 2, 3, 4$. The pooled measured/predicted ratio per cell is 0.906–1.022, and every pooled cell is within 4 empirical SE.
   - The registered Poisson rule (H-A) **failed** on 25 rows: ordered counts are clustered, and the rule ignored that.
   - The amended rule (H-A′, empirical per-target variance, same targets) fails on **one** row: $N = 15$, standard line, $z = -4.35$. Over all 87 rows, z has mean -0.28 and sd 1.24.
   - Amendment 3 fixes a population mismatch: targets are never O or $T = \left(0, \sqrt{b}\right)$, so the matching expectation is $(|F|^m - C_O - C_T)/(\#E - 2)$, with C_O and C_T counted exactly. Scored against it (H-A″), the same single row fails ($z = -4.35$), and z has mean -0.18 and sd 1.26.
   - No bias is indicated. The residual overdispersion is a limit of the sampling model, not of the identity.
2. **Twisted line (theorem).** Let $N = dn$, $T = \mathrm{Tr}_{N/d}$ and $c = \sqrt{b}$. For $x = uy$ with $y \in \mathbb{F}_{2^d}$:
   - $\mathrm{Tr}_N(c/x) = \mathrm{Tr}_d(T(c/u)/y)$;
   - $\mathrm{Tr}_N(x) = \mathrm{Tr}_d(T(u)y)$.

   So when $T(u) = T(c/u) = 0$, the lift condition on the $\mathbb{F}_{2^d}$-line $u\mathbb{F}_{2^d}$ is the constant $\mathrm{Tr}(a)$: all-lift if $\mathrm{Tr}(a) = 0$, all-twist if $\mathrm{Tr}(a) = 1$. The line is an $\mathbb{F}_{2^d}$-subspace, so Gaudry–Diem descent applies with the same monomial support.

   Proposition 3 (corrected in Amendment 3). Existence of u is a projective condition on $P(\ker T)$, with $q = 2^d$:
   - $n = 2$: $\ker T = \mathbb{F}_q$, so u exists iff $c \in \mathbb{F}_q$, and then every $u \in \mathbb{F}_q^*$ works;
   - $n = 3$: $T(c/u) = T\left(cu^q u^{q^2}\right)/N(u)$ is a binary quadratic form on $\ker T \cong \mathbb{F}_q^2$. Its zero set in P^1 has 0, 1 or 2 points, or all $q + 1$ if the form vanishes. So the count is in $\{0, q-1, 2(q-1), q^2-1\}$;
   - $n = 4$: a plane cubic, so the count is $(q-1)$ times its points.

   The first version claimed $\{0, 2(q-1)\}$ at $n = 3$. That is wrong: Codex review found $N = 15$, $d = 5$, $b = 4$ with count 31, the single-root case.

   Exact computation, corrected rule (H-B′): pass.
   - Exhaustive census over every b at $(N, d) = (12, 6), (15, 5), (18, 6), (18, 9), (16, 4)$, plus 4096 sampled b at $(21, 7), (20, 5)$: 0 violations.
   - At $N = 15$ the counts are 0 (15,376 of b), 31 (1,023 $\approx 1/q$) and 62 (16,368).
   - At $n = 4$, u existed for every b checked (minimum 150 at $N = 16$).

   Line curves (H-B as registered, also H-B′): pass on 72 of 72.
   - With $\mathrm{Tr}(a) = 0$, every twisted line lifts 100 %.
   - With $\mathrm{Tr}(a) = 1$, every one lifts 0 %.

- $n = 2$: no u on 32 of 32 random curves.
- $n = 3$: 11 of 24 curves had a u; none was degenerate.
- $n = 4$: counts are multiples of $2^d - 1$ (180–1116), and u exists on 16 of 16 curves.

3. **Real parameters, c2pnb (exact computation).** For the ANSI X9.62 c2pnb curves ($N = 16k$, $a, b \in \mathbb{F}_{2^{16}}$), u is taken in $\mathbb{F}_{2^{16}}$, so $u\mathbb{F}_Q$ stays π_{16}-stable.
   - **c2pnb208w1** and **c2pnb368w1** have $\mathrm{Tr}(a) = 0$. They have 180 and 330 valid u. 200/200 sampled twisted elements lift, against 103/200 and 95/200 on the standard base $\mathbb{F}_Q$: **applicable**.
   - **c2pnb176v1, 272w1, 304w1** have $\mathrm{Tr}(a) = 1$. The lines lift 0/200: **inapplicable**. The trace homomorphism puts the whole line on the twist.
   - The native check reproduces the historical Python output on every deterministic field: $\mathrm{Tr}(a)$, $\mathbb{F}_{2^{16}}$ membership, the kernel size and the u-counts.

4. **Cost (model output, not a measurement).** Natively, in closed form: the 2LP optimum $T^* \propto \sqrt{N^3/p}$ with $N' = 2N$ and $p' = 16p$ changes the margin by $+0.50$ bits.

   The upstream C1 Gaudry–Diem $n = 4$ model (Python, crypto-autoresearcher) was run once with only those two inputs changed. Its output is quoted as historical evidence and not regenerated:
   - Under double large primes, for the applicable rows at $c_{\text{trial}} = 2^{27}$ and 2^{30}:
     - time, memory model (a) and mesh-a: **+0.50 to +0.78 bits**;
     - memory model (b), whose storage grows with the base: -0.51 to $+0.57$ bits.
   - The time figure matches the analytic $T^* \propto \sqrt{N^3/p}$, factor $2^{-1/2}$.
   - The line always contains a standard-size sub-base: keep only half its points and the relation rate equals the standard one. So a solver can fall back, and the option is weakly dominant (an argument, not a measurement).
   - Under basic Gaudry the model is LA-dominated, and the doubled base costs -2 bits.
   - The model holds c_{trial} fixed, an assumption. A measured improvement needs the upstream C1 toy ladder run with both bases.

5. **Prime degree, repulsion (measured, confirmatory at $N = 11$).** Exact unpruned counts were taken of all-lift k-dimensional subspaces, normalised by the exact expectation for a uniformly random set of the same size. The nulls are paired one-to-one by density.

   At $k = 4$, curves against nulls:

N	curves (mean ± sd)	nulls (mean ± sd)	one-sided permutation p
9 (26 vs 26)	0.683 ± 0.166	1.021 ± 0.262	$2 \cdot 10^{-5}$
11 (26 vs 26)	0.921 ± 0.018	1.027 ± 0.053	$2 \cdot 10^{-5}$ — **confirmed**
12 (6 vs 6, composite)	0.989 ± 0.001	0.991 ± 0.020	0.51
13 (6 vs 6)	0.981 ± 0.005	0.994 ± 0.010	0.019

   Lift sets contain **fewer** all-lift 4-subspaces than random sets, and their counts are far less variable. The deficit shrinks with N at fixed k. Exact maxima at prime $N \le 13$: the curve median kmax is at most the null median (4 vs 4 at $N = 9$, 4 vs 5 at $N = 11$, 5 vs 5 at $N = 13$). At composite $N = 12$ the koblitz curves reach 6 against 5, through the subfield construction. The planted subspaces were recovered on 100 % of instances.

   Above $N = 13$ the budgeted search recovered **none** of the planted subspaces, so those kmax values (all 5–6) are "found under budget" only. The subfield construction at composite N shows what the budgeted search misses: dimensions 8–11 at $N = 16$–22.

6. **ECC2K-130 ($N = 131$, prime): no gain (theorem plus toy evidence).**
   - There is no subfield, so no twisted line exists.
   - Weil: an all-lift subspace satisfies $\sum_{u \in V^\perp} K(c(1+u)) = \pm 2^N$ with $|K| \le 2^{N/2+1}$. Hence its dimension is $\le N/2 + 1$.
   - Random-set heuristic: the largest all-lift dimension is about 10 at $N = 131$, far below the 33–44 that index calculus needs.
   - The toy data show repulsion, not attraction.
   - **Not established:** that no algebraic all-lift family exists at prime N. The search cannot rule out needles above $N = 13$.

1.3. Figures

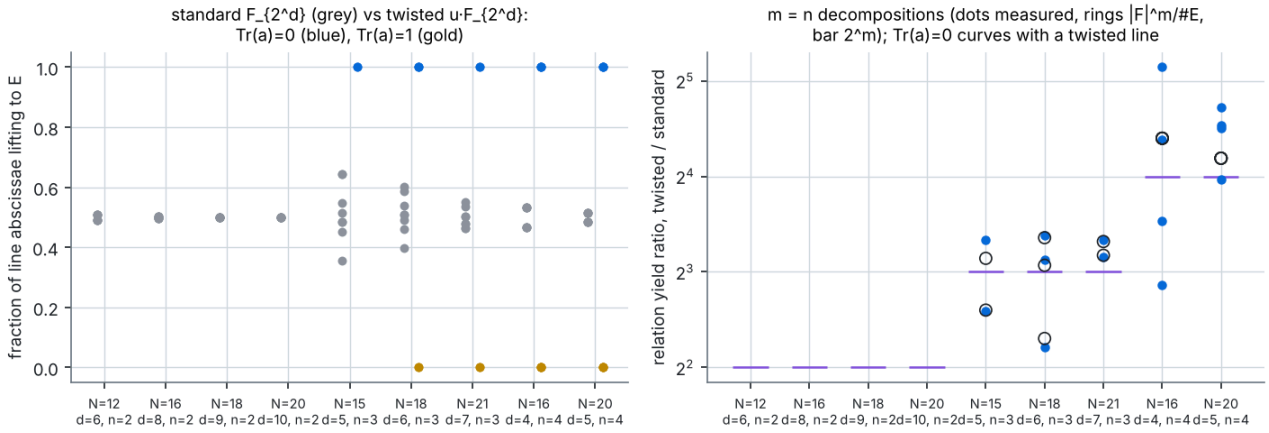


Figure 1: Composite degree. Left: lift fraction of the standard and twisted lines. Right: twisted/standard relation yield, measured (dots) and predicted (rings).

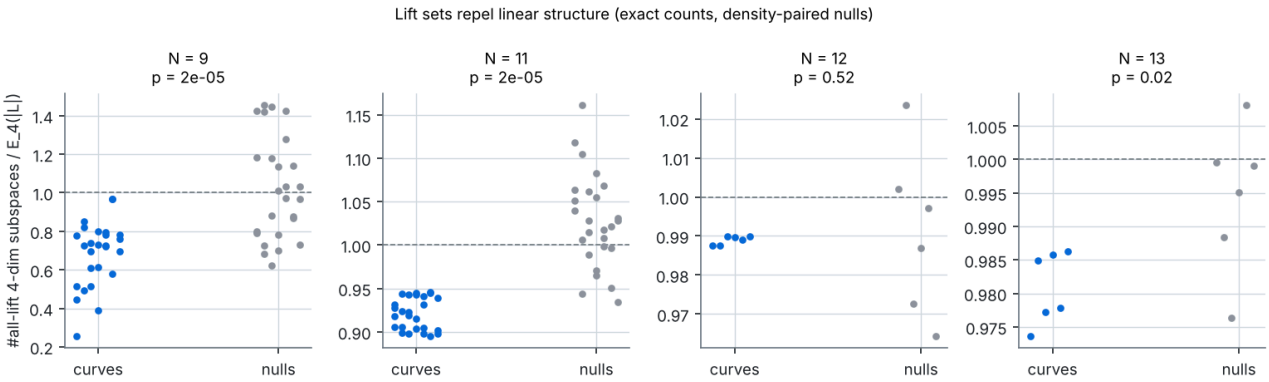


Figure 2: Prime degree: normalised all-lift 4-subspace counts, curves against density-paired nulls.

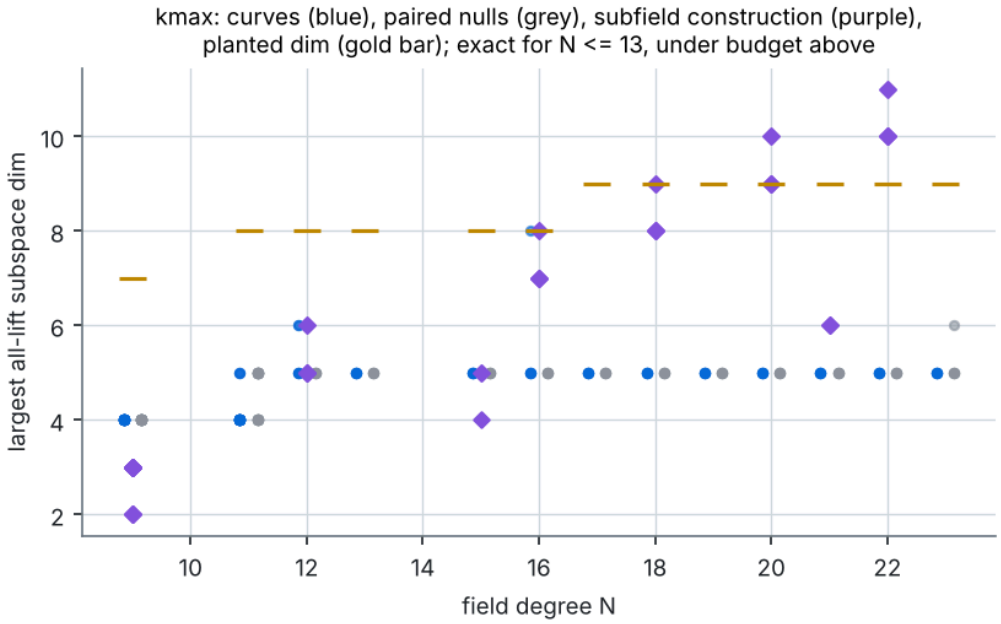


Figure 3: Largest all-lift subspace by N. Blue: curves; grey: nulls; purple: subfield construction; gold: planted dimension.

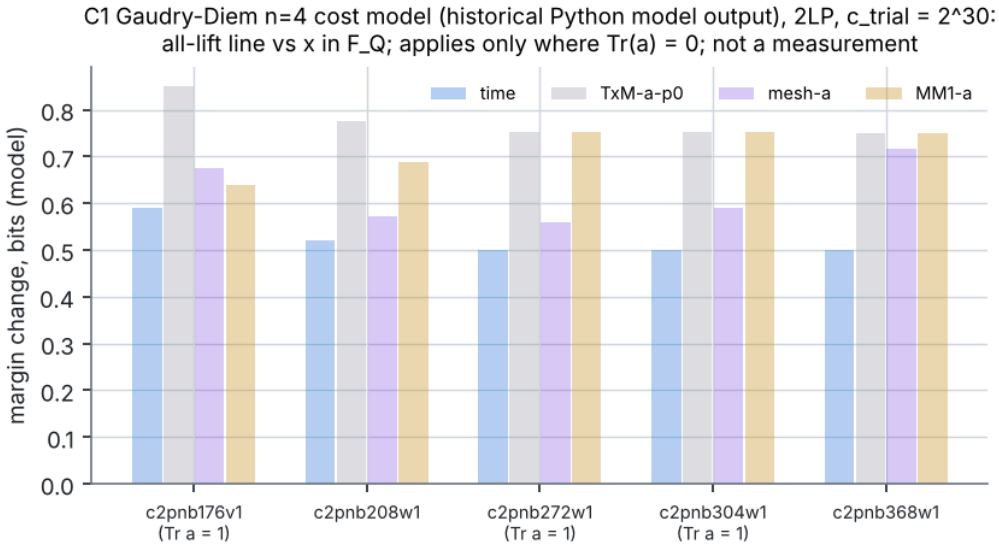


Figure 4: C1 cost-model margin change (model output). Faded bars: $\mathrm{Tr}(a) = 1$, not applicable.

1.4. The trace homomorphism: where it acts and what it buys

- **As a sum constraint (closed).** The Kummer map $P \mapsto \mathrm{Tr}(x_P) + \mathrm{Tr}(a)$ is the Kosters–Yeo trace equation: one linear equation, falling at degree 2. Any homomorphic filter is capped at $\log_2$ of the cofactor (2 bits on ECC2K-130).
- **On points, $\mathbf{Tr}_{q^k/q}$ (closed).** Trace-zero and extension embeddings either kill the rational line or enlarge the problem.
- **As a membership condition on the base (this study).**
  - $\mathrm{Tr}_{N/d}$ makes the lift condition constant on a twisted subfield line. That is worth 2^n per trial at composite degree with $\mathrm{Tr}(a) = 0$, and nothing at prime degree.
  - It also decides applicability: on c2pnb176v1, 272w1 and 304w1 the same construction lands on the twist.

1.5. Lean

`lean/LiftDense.lean` is a Lake project on the repository Mathlib pin.

- `lake build` is clean.
- The repository proof gate passes: 11 theorems, all on `propext`, `Classical.choice` and `Quot.sound`, with no `sorry` and no `native_decide`.

Proved, over any finite field M of characteristic 2 and any intermediate field L:

- Lemma 1, the trace tower (`trace_mul_algebraMap`).
- Theorem 2 in trace form (`twisted_line_trace_const`).
- $\mathrm{Tr}(z^2) = \mathrm{Tr}(z)$ (`trace_sq`).
- The Artin–Schreier criterion $\exists z, z^2 + z = w \iff \mathrm{Tr}(w) = 0$ (`artin_schreier`). The range of $z \mapsto z^2 + z$ lies in $\ker \mathrm{Tr}$, and both have dimension $n - 1$.
- The curve lift criterion (`lifts_iff`).
- Theorem 2 in lift form (`twisted_line_lifts`): for $y \in L^*$, yu lifts to E_{a,c^2} iff $\mathrm{Tr}(a) = 0$.

Still open, recorded in `PROOF_MAP.md`:

- Proposition 3, the u-count classification. It is checked exhaustively by census.
- The yield identity.
- The Weil cap. This needs the Kloosterman bound, which is not in Mathlib.

1.6. Not claimed

- No ECDLP was solved and no ECDLP cost was measured, so there is no leaderboard, benchmark or rho row.
- The c2pnb figure is a model delta.
- The repulsion is toy-size evidence ($N \le 13$, $k = 4$). Its mechanism (presumably Kloosterman-sum rigidity) is not established.

1.7. Next

1. Run the upstream C1 toy ladder ($E/\mathbb{F}_{2^4}$ lifted to $\mathbb{F}_{2^{4k}}$) with both bases. This measures the relation rate per trial and c_{trial}. It decides whether the +0.5 bit is real for c2pnb208w1 and 368w1.
2. Apply the same line to the Oakley EC2N quintic-tower lane ($n = 5$, $2^5 = 32$ per trial), after checking $\mathrm{Tr}(a)$.
3. Derive the repulsion constant at $k = 4$ from the fourth moment of Kloosterman sums, and test it against $N = 15$ counts.